

1.3 Summation of series

Further Pure 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Use finite-sum identities for r , r^2 , r^3 and partial fractions for telescoping sums. Induction is useful to prove a proposed result: verify base case, assume $n=k$, then derive $n=k+1$ explicitly. An infinite geometric sum requires $\text{abs}(r) < 1$; never apply it to divergent series.

Worked example: follow the reasoning

Σ from $r=1$ to n of $(1/r - 1/(r+1))$ telescopes to $1 - 1/(n+1)$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Find Σ from $r=1$ to n of $1/[r(r+1)]$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Partial fractions $\frac{1}{r(r+1)} = \frac{1}{r} - \frac{1}{(r+1)}$ [1]; cancellation [1]; sum = $1 - \frac{1}{(n+1)} = \frac{n}{(n+1)}$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.